

Soumadeep Saha, PhD

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Summary

My current research at ANITI explores techniques to better understand *reasoning with large language models*. My PhD research centered on enhancing deep learning systems with the strengths of symbolic AI techniques for greater logical coherence and domain constraint adherence. While my primary focus is *natural language processing*, I actively collaborate across domains including vision, medicine, biology, astrophysics, and business applications.

Education

Indian Statistical Institute, Kolkata

KOLKATA, INDIA

PhD (Computer Science)

2020–2025

☐ Supervisor: **Prof. Utpal Garain**

☐ Dissertation: “**Domain Obedient Deep Learning**”.

Indian Institute of Science Education and Research, Kolkata

KOLKATA, INDIA

Integrated BS & MS in Physics (Minor in Mathematics)

2015–2020

☐ Master’s dissertation: “**Towards Robust Deep Learning Systems**”.

Selected Works [8]

☐ TAPIOCA: Why Task-Aware Pruning Improves OOD model Capability

Accepted to NeurIPS 2026

K. Sharma, O. Naim, **S. Saha**, V. Jain, A. Chadha, N. Asher

10.48550/arXiv.2605.14738

☐ BODHI: Do LLMs Branch Out and Discover Heterogeneous Inferences?

Preprint

S. Saha, K. Sharma, A. Chaturvedi, N. Asher

10.48550/arXiv.2608.02867

☐ The Synergistic Integration of Access Control Management and Large Language Model Agents: A Survey

Preprint

F. Periti, **S. Saha**

10.36227/techrxiv.177075184.43798334/v1

☐ KisMATH: Do LLMs Have Knowledge of Implicit Structures in Mathematical Reasoning?

TACL (2026), 14: 1308-1328

S. Saha, A. Chaturvedi, S. Saha, U. Garain, N. Asher

☐ sudoLLM: On Multi-role Alignment of Language Models

EMNLP 2025 (findings)

S. Saha, A. Chaturvedi, J. Mahapatra, U. Garain

☐ Language Models are Crossword Solvers

NAACL 2025 (main)

S. Saha, S. Chakraborty, S. Saha, U. Garain

☐ Analyzing Semantic Faithfulness of Language Models via Input Intervention on Conversational Question Answering

Computational Linguistics (2024), 50(1): 119-155

A. Chaturvedi, S. Bhar, **S. Saha**, U. Garain, N. Asher

☐ On Measuring Intrinsic Causal Attributions in Deep Neural Networks

CLear 2025

S. Saha, D. V. Rathore, **S. Saha**, D. Doermann, U. Garain.

☐ VALUED - Vision and Logical Understanding Evaluation Dataset

DMLR (2024), (13):1-18.

S. Saha, S. Saha, U. Garain

☐ MedTric : A clinically applicable metric for evaluation of multi-label computational diagnostic systems

PLOS One (2023), 18(8): e0283895

S. Saha, U. Garain, A. Ukil, A. Pal, S. Khandelwal.

Patents

☐ Method and System for Contradiction Avoided Learning for Multi-Class Multi-Label Classification

US Patent - US12038949B2

S. Saha, U. Garain, A. Ukil, A. Pal

Granted Jul. 2024

☐ Method and System for Evaluating Clinical Efficiency of Multi-Label Multi-Class Computational Diagnostic Models

Pending

U. Garain, **S. Saha**, A. Ukil, T. Deb, S. Richa A. Pal, S. Khandelwal

Filed Sep. 2022.

Grants, Fellowships & Awards

OpenAI Grant

Jan '25

Awarded \$5000 in computational credits from **OpenAI** under their *Researcher Access Program* to support my work on **LLM alignment**.

Helmholtz Visiting Researcher Grant

Jul '24 – Sep '24

Awarded the Helmholtz Information and Data Science Academy (HIDA) *visiting researcher grant* to work at the Institute of Aerospace Medicine, DLR (German Aerospace Center). Conducted research on improving AI-based retinal diagnostics for spaceflight applications.

KVPY & NTSE

2015, 2013

Awarded the prestigious Kishore Vigyan Pratoshan Yojna (**KVPY**) fellowship from the Department of Science and Technology, Govt. of India, and recipient of the National Talent Search Examination (**NTSE**) Scholarship from NCERT, Govt. of India.

Experience

Alleima

Mar '24 – May '24

Developed a custom end-to-end computer vision-based solution for assembly-line object detection/tracking, enabling autonomous logging of crucial data with existing infrastructure.

TCS Research

Nov '21 – Jul '22

Conducted research as part of a team on ECG-based diagnosis of cardiovascular diseases. Independently identified key gaps in the literature and developed innovative, state-of-the-art solutions to address them, resulting in two patents, and publications.

Talks and Presentations

- ❑ Organized a 4-day workshop for DataLab, **Capital One**, Bangalore (2023).
- ❑ Subject matter expert for Be10x; designed MOOCs, and delivered live seminars to 500+ participants (2024).
- ❑ TA for Natural Language Processing course at ISI Kolkata (2022–2025).
- ❑ Instructor at the Winter School of Deep Learning (WSDL), ISI Kolkata (2021–2024).
- ❑ Instructor for the Comprehensive Course on Business Analytics, ISI Kolkata (2022).
- ❑ Presented my work on Logically Coherent Deep Learning at Amazon Research Days (ARD '22).
- ❑ Presented my work on domain-obedient self-supervision at W3PHIAI, AAAI 2024, Vancouver, Canada.
- ❑ Presented my work on constrained LLM inference at NAACL 2025, Albuquerque, USA.

Technical Skills

- ❑ **Languages & Frameworks:** Python (PyTorch, HuggingFace), C++, bash.
- ❑ **Infrastructure & Compute:** Multi-node and multi-GPU distributed (FSDP, DeepSpeed) training (RLVR, SFT) on GPU clusters.
- ❑ **LLMs & Alignment:** RLVR, RLHF, DPO, PEFT/Q-LoRA, RAG, In-context Learning.
- ❑ **Computer Vision:** VLMs, Vision Transformers (ViTs), CNNs, Object Detection & Segmentation.

Interests

- ❑ **Robotics/DIY**—Active interest in 3D printing, homelabbing, robotics, electronics, etc. I have conducted introductory workshops on robotics and was the Secretary of the Robotics and Astronomy Club at IISER Kolkata.
 - ❑ **Sports**—Represented my college in national-level sports meets in basketball and volleyball. Played in my state's Senior Division Men's Basketball League.
 - ❑ **Music**—Classically trained pianist.
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